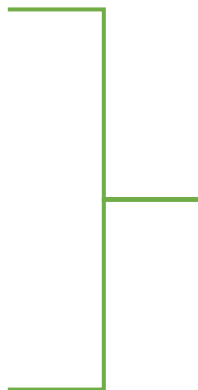


Coulometry

- In **potentiometry**, the potential of an electrochemical cell under static conditions is used to determine an analyte's concentration.
- Dynamic electrochemical methods,
 - **Coulometry**,
 - **Voltammetry**, and
 - **Amperometry**,

current passes through the electrochemical cell

✓ Also are important analytical techniques.

COULOMETRY:

- An **electrochemical method** in which the current required to **oxidize or reduce the analyte is measured**.
- Analyte is **quantitatively oxidized or reduced at the working** electrode or reacts quantitatively with a reagent generated at the working electrode.

- **There are two forms of coulometry:**
 - **Controlled-potential coulometry**, in which a constant potential apply to the electrochemical cell, and
 - **Controlled-current coulometry**, in which a constant current pass through the electrochemical cell.
- The **total charge, Q** , in coulombs, passed during an electrolysis is related to the **absolute amount of analyte by Faraday's law**.

Faraday's law: The current or charge passed in a redox reaction is proportional to the moles of the reaction's reactants and products.

$$Q = nFN \text{(1)}$$

- **Where:** n is the number of electrons transferred per mole of analyte,
 - F is Faraday's constant (96487 C mol^{-1}), and
 - N is the moles of analyte.
- A coulomb is also equivalent to an **A×s**; thus, for a constant current, i, the charge is given as,

$$Q = i \times t \text{(2)}$$

- If current varies with time, as **it does in controlled potential Coulometry**, then the total charge is given by,

$$Q = \int_{t=0}^{t=t_e} i(t) dt \dots (3)$$

- In coulometry, **we monitor current as a function of time** and use either equation 2 or 3 to calculate Q.
- Knowing the total charge, we then use equation 1 to determine the **moles of analyte**.

- To obtain an accurate value for N , therefore, **all the current must result in the analyte's oxidation or reduction.**
- NB: coulometry requires 100% current efficiency

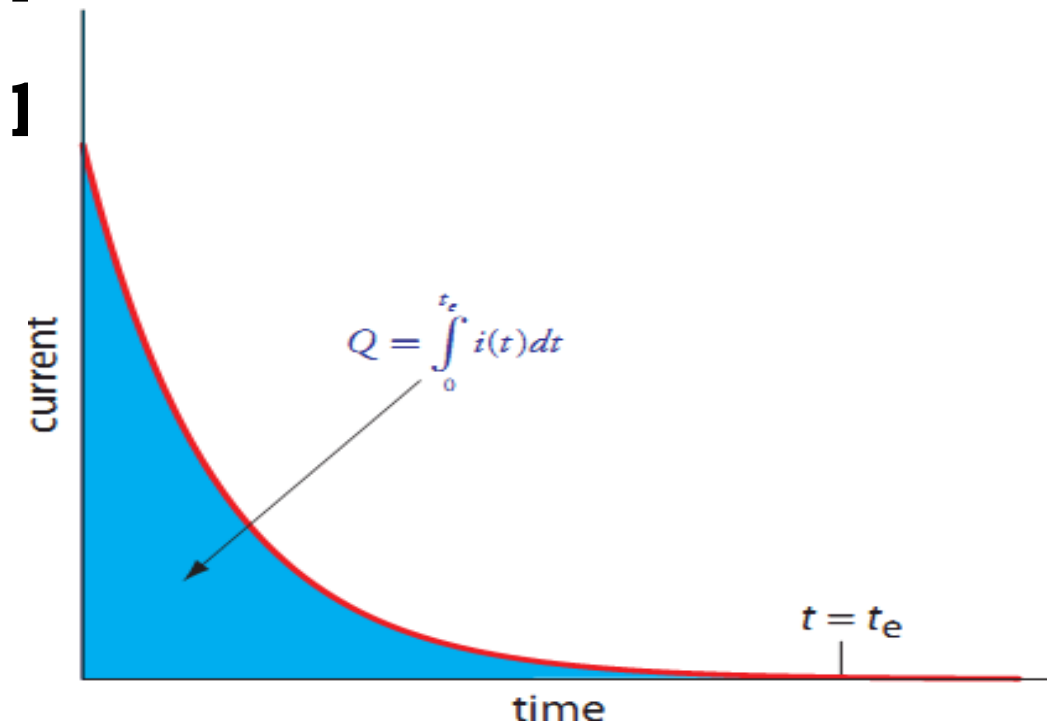
Current efficiency: is the percentage of current that actually leads to the analyte's oxidation or reduction.

Controlled-potential coulometry

- The easiest method for ensuring **100% current efficiency** is to **maintain the working electrode at a constant potential**.
 - allows for the analyte's quantitative oxidation or reduction, without simultaneously oxidizing or reducing an interfering species.
- The **current flowing through an electrochemical cell** under a **constant potential** is proportional to the **analyte's concentration**.

- As electrolysis progresses the **analyte's concentration decreases**, as **does the current**.
- The resulting **current-versus-time profile** for **controlled-potential coulometry**, which also is known as

is shown



✓ *The integrated area under the curve, shown in blue, is the total charge(Q)*

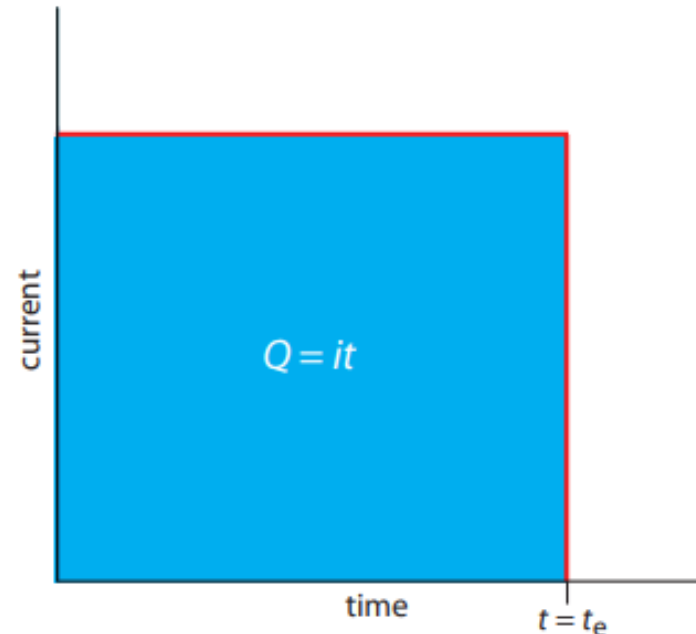
Instrumentation:

- The potential in controlled-potential coulometry is set using a **three-electrode potentiostat**.
 - Two types of working electrodes are commonly used: **a Pt electrode and Hg electrode**.
 - A **saturated calomel or Ag/AgCl** electrode serves as the reference electrode.

- The other essential feature is **a means of determining the total charge passed during electrolysis.**
- **(1)** *One method is to monitor the current as a function of time and determine the area under the curve*
- **(2).** Modern instruments **use electronic integration.**
 - *The total charge at the end of the electrolysis then can be read directly from a digital readout* or from a plot of charge versus time

Controlled-Current Coulometry

- A second approach to coulometry is to use a **constant current** in place of a constant potential.
- **Controlled-current coulometry**, also known as *amperostatic coulometry* or *coulometric titrimetry*,



Current versus time for a controlled-current coulometric analysis.

- It has **two advantages** over controlled-potential coulometry.
- First, using a constant current makes for **a more rapid analysis** since the current does not decrease over time....**less than 10 min.**
- Second, with a constant current the **total charge is simply the product of current and time.....** integrating the current–time curve, therefore, is not necessary.

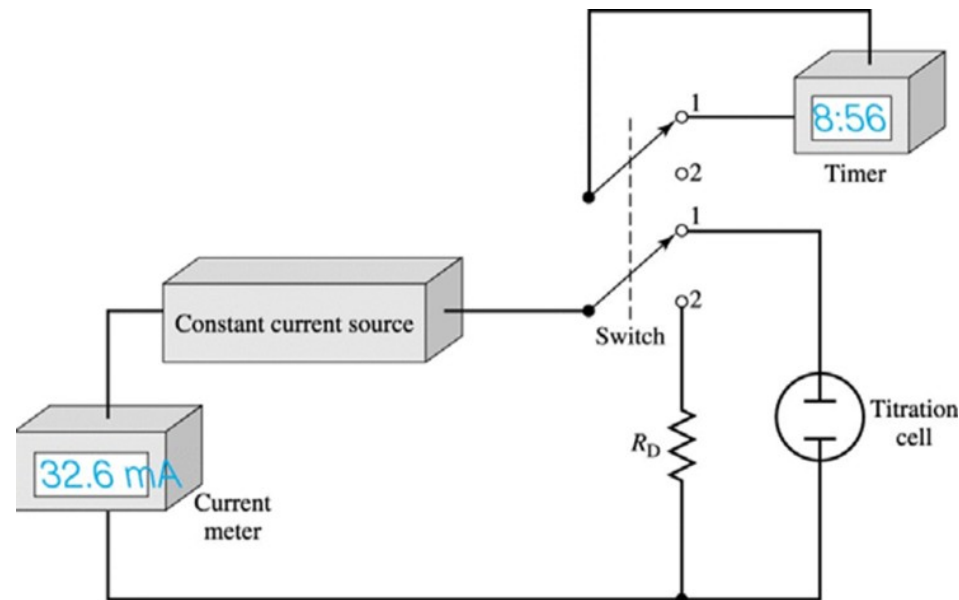
$$Q = it$$

- *There is a need for a method of determining when the analyte has been exhaustively electrolyzed.*
- In controlled-potential coulometry this is signaled by a decrease in the current to a constant background or residual current.
- Like titrimetry, visual indicators, and potentiometric measurements, may be used to signal the end point of a controlled-current coulometric analysis.

Instrumentation:

- **Controlled-current coulometry** normally is carried out using **a galvanostat** and an electrochemical cell consisting of a **working electrode and a counter electrode**.
- The other necessary instrumental component for controlled-current coulometry is
 - **an accurate clock** for measuring the electrolysis time, t_e , and
 - a **switch** for starting and stopping the electrolysis.

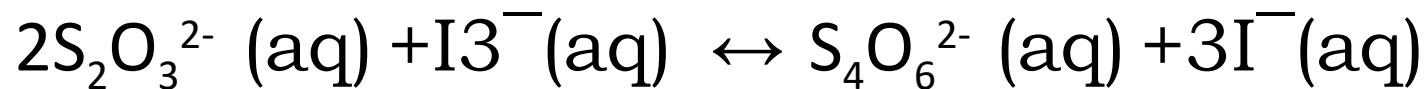
- **Digital clocks** provide a more accurate measurement of time.
- A **switch controls** the flow of current and the clock, so that an accurate determination of the **electrolysis time** is possible.



Coulometric calculation example

- To determine the purity of a sample of $\text{Na}_2\text{S}_2\text{O}_3$ a sample is titrated coulometrically using I^- as a mediator and I_3^- as the titrant. A sample weighing 0.1342 g is transferred to a 100-mL volumetric flask and diluted to volume with distilled water. A 10.00-mL portion is transferred to an electrochemical cell along with 25 mL of 1 M KI, 75 mL of a pH 7.0 phosphate buffer, and several drops of a starch indicator solution. Electrolysis at a constant current of 36.45 mA requires 221.8 s to reach the starch indicator endpoint. Determine the sample's purity.

Solution:



- The oxidation of $2\text{S}_2\text{O}_3^{2-}$ to $\text{S}_4\text{O}_6^{2-}$ requires one electron per $\text{S}_2\text{O}_3^{2-}$

$$N = \frac{it}{nF} = \frac{(0.03645\text{A})(221.8\text{S})}{\left(\frac{1\text{mol } e}{\text{mol Na}_2\text{SO}_3}\right)(96487\text{C})} = 8.379 \times 10^{-5} \text{ mol Na}_2\text{S}_2\text{O}_3$$

However, $n = \frac{\text{given mass}(m)}{\text{mwt}(M)}$

$= m = n \times M$

$= 8.379 \times 10^{-5} \text{ mol} \times 158.1\text{g} = 0.01325\text{g Na}_2\text{SO}_3$

- This is the amount of **Na₂SO₃** in a 10.00 mL portion of a 100 mL sample; thus, there are 0.1325 grams of **Na₂SO₃** in the original sample.
- The sample's purity, therefore, is

$$\frac{0.01325 \text{ g Na}_2\text{SO}_3}{0.1342 \text{ g sample}} \times 100\% = 98.73\% \text{ w/w Na}_2\text{SO}_3$$

Thank you

Quiz (5%)

- In coulometric analysis, a constant current of 0.8 A is used to deposit copper at the cathode and oxygen at the anode of an electrolytic cell. Calculate the number of grams of each product formed in 15.2min, assuming no other redox reaction

- The two half-reactions are:

